

Lower Log-Scale-Integrated Permutation Entropy During a Selected Propofol-Sedation Acquisition: A Paired Reanalysis of OpenNeuro ds005620

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Authorial disclosure: This manuscript was AI-generated in the fictional authorial voice of Dr. Sabine Laghari. It has not been reviewed by a human.

Abstract

We compared one selected eyes-closed awake acquisition with the one-minute `sed2` acquisition immediately preceding the first awakening attempt in 20 paired participants from OpenNeuro ds005620 version 1.0.0 [2, 1]. The supplied protocol designated log-scale-integrated permutation entropy as the primary endpoint. The analysis used the first 60 seconds of each acquisition, 62 EEG channels, four-second nonoverlapping windows, permutation order 3 with unit delay and stable index-based tie resolution, and nonoverlapping block-mean coarse-graining at scales 1, 2, 4, and 8. Entropy was first median-aggregated across retained windows within channel. Per-channel values were integrated by normalized trapezoidal quadrature over $\log_2$ scale, and the participant-condition endpoint was the median of these areas across channels.

Mean endpoint values were 0.937074 during wakefulness and 0.920556 during sedation. The paired mean difference, sedated minus awake, was -0.016518 (paired $d_z = -0.458$); 14 of 20 differences were negative. Exhaustive enumeration of all 2^{20} sign assignments gave a two-sided sign-flip value of $p = 0.054569$. A 10,000-resample participant-paired percentile bootstrap interval was $[-0.032088, -0.001146]$. The enumeration is computationally exact, but its inferential exactness requires sign exchangeability, which was not established by randomized acquisition labels.

Point estimates were also negative with two-second windows (-0.018293), in posterior channels (-0.005401), and in frontal channels (-0.024620). Thus, the protocol's literal sign-direction criterion was met descriptively. However, the posterior interval included zero and increases, and no smallest relevant effect, uncertainty requirement, multiplicity rule, or confirmatory decision threshold was specified.

A separately implemented post-run verifier recomputed the primary difference and sign-flip result from the frozen participant table, checked completion, uniqueness, finiteness, duration, channel count, physical scaling, source hashes, and consistency with stored scale columns, and reproduced a synthetic estimator sanity check. This verification establishes internal consistency of the frozen artifacts, not independent scientific replication.

The endpoint is a weighted average of entropy level over selected log-spaced scales. A lower value does not by itself demonstrate reduced scale-profile breadth, persistence, slope, curvature, or entropy at every scale. The nonrandomized acquisition contrast also does not identify a propofol-specific causal effect. No conclusion

about dreaming, awareness, phenomenal consciousness, or absence of experience follows from this acquisition-state comparison.

Introduction

Permutation entropy summarizes the relative frequencies of ordinal patterns in a time series ^[3]. Its value is invariant under strictly monotone transformations because such transformations preserve or consistently relabel ordinal patterns. Non-strict transformations, quantization, and coarse-graining can create ties, making the result dependent on the implementation's tie rule.

Applying entropy estimation to coarse-grained signals provides descriptions at multiple temporal resolutions. Costa, Goldberger, and Peng motivated multiscale analysis of physiological time series, although their framework does not independently validate the present permutation-entropy estimator, chosen scale grid, or scalar integration rule ^[4].

An integrated entropy score must be distinguished from the shape or “breadth” of a scale profile. Two parallel profiles separated by a constant have the same scale dependence but different integrated values. Likewise, an area difference can arise predominantly at one scale or despite crossing profiles. A lower integrated score therefore establishes a lower weighted average under the selected quadrature, not necessarily diminished persistence or breadth across scales.

Previous work reported lower multidimensional spontaneous-EEG complexity during propofol general anesthesia ^[5]. That study provides background motivation rather than direct replication: its multidimensional estimator and general-anesthesia condition are not interchangeable with the present univariate ordinal statistic and selected sedation acquisition.

Spontaneous permutation entropy must also be distinguished from perturbational complexity. The perturbational complexity index characterizes the spatiotemporal cortical response to external perturbation ^[6]. Passive permutation entropy instead describes an ongoing signal and can vary with autocorrelation, oscillatory composition, filtering, noise mixture, artifacts, and segment selection. Findings about one measure do not validate the other.

The source dataset was collected for a repeated-awakening investigation of dreaming during propofol sedation ^[1]. The supplied protocol and aggregate record state that the available BIDS event tables did not expose the companion study's dream-report labels. No experience/no-experience or dream/no-dream labels were inferred ^[7]. The present target is consequently an acquisition-state-associated EEG statistic, not conscious experience.

Hypothesis and Registered Analysis

The supplied protocol stated:

Within subjects, propofol sedation reduces permutation-entropy breadth across temporal scales relative to eyes-closed wakefulness, and the direction survives window-length and regional sensitivity analyses.

The README calls this the registered hypothesis and identifies `multiscale_entropy_auc` as the registered primary endpoint. The configuration and complete analysis source specify the associated computation. However, the supplied package does not include an independently time-stamped registration or version history establishing that the hypothesis, segment rule, regions, preprocessing choices, and estimator settings were fixed before outcome access. “Registered” is therefore used here as a protocol designation, not as a claim of independently verified temporal preregistration.

The estimand was

$$d_i = B_{i,\text{sedated}} - B_{i,\text{awake}},$$

where B_{ic} is the participant-condition log-scale-integrated permutation-entropy score. A negative $\bar{d}$ means that this score was lower during the selected sedated acquisition.

The original wording requires construct and causal qualification. The endpoint is an integrated entropy level rather than a unique measure of scale-profile breadth, and the nonrandomized acquisitions do not isolate a propofol-specific causal effect. The directional sensitivity criterion is operationally identifiable—negative point estimates in the two-second, posterior, and frontal analyses—but it does not require meaningful magnitude or uncertainty control.

Evidential role	Analysis	Interpretation
Protocol-designated primary	Whole-scalp integrated entropy, four-second windows	Primary stored contrast; temporal preregistration not independently established
Pipeline pilot	Four configuration-designated participants	Used by the <code>--pilot</code> mode; pilot-only results were not supplied
Full analysis	All 20 configured participants, including the four pilots	Pooled paired result; not a held-out-only confirmation
Sensitivity	Two-second windows, posterior channels, frontal channels	Directional comparisons with no joint inferential rule
Exploratory	Single-scale permutation entropy	Helps assess a general entropy-level explanation
Contextual	Spectral exponent and relative alpha power	Describes accompanying spectral differences
Estimator check	Synthetic white noise versus periodic sequence	Implementation sanity check, not a biological control

No alpha level, smallest effect of interest, multiplicity procedure, regional uncertainty criterion, or binary success rule was specified in the supplied protocol or configuration. We therefore report estimates and procedure-specific uncertainty without declaring confirmatory support.

Methods

Dataset, participants, and acquisition selection

The source was OpenNeuro ds005620 version 1.0.0, DOI 10.18112/openneuro.ds005620.v1.0.0 [2]. Its acquisition context is described in the companion publication [1].

The sanitized configuration specifies 20 analyzed participants and one excluded dataset participant. Participant identifiers, the public participant table, and the exclusion rationale remain withheld from the manuscript package. This figure-rich revision used the local table only to generate anonymized paired and scale-profile graphics. All 20 configured participants completed both conditions without recorded analysis failure. Four were designated for pipeline-pilot execution; the full analysis included all 20.

The selected acquisitions were:

- Awake: `task-awake_acq-EC`
- Sedated: `task-sed2_acq-rest_run-1`

The protocol describes `sed2` as the one-minute recording immediately before the first awakening attempt. The code analyzes exactly the first 60 seconds of both selected acquisitions, thereby matching duration before preprocessing. The awake file may be longer, but later samples are not analyzed [7].

Dose, behavioral responsiveness, physiological covariates, and a quantitative sedation-depth definition were not included in the supplied evidence. Acquisition-order randomization was not documented. Pairing controls stable participant characteristics but does not remove time-varying differences involving order, vigilance, fatigue, recording context, preceding events, or pharmacological state.

Data loading, physical scaling, and channel selection

The source reads BrainVision recordings only when the header declares multiplexed orientation and IEEE float32 binary data. The binary array is reshaped into samples by channels, and the first 60 seconds at the header-derived source sampling rate are selected.

Channels named `VEOG`, `HEOG`, `EMG`, `ECG`, or `EOG`, irrespective of case, are excluded. All analyzed recordings passed the verifier's 62-EEG-channel check.

Each retained channel is converted to microvolts using its BrainVision resolution and unit. Supported units are μV , uV , mV , and V , with the corresponding conversion factors. The verifier found minimum and maximum scaling factors of $0.1 \mu\text{V}$ in every analyzed participant-condition recording.

Filtering, downsampling, and referencing

The configured target sampling rate was 250 Hz. The source rate was required to be evenly divisible by the integer downsampling factor; its numerical value was read from each BrainVision header rather than fixed in the sanitized configuration.

For each full 60-second channel:

1. The temporal mean was removed.
2. A real Fourier transform was computed.
3. Frequencies at or below 80 Hz retained unit gain.

4. Frequencies from 80 to 100 Hz received a half-cosine transition:

$$g(f) = \frac{1}{2} \left[1 + \cos \left(\pi \frac{f - 80}{100 - 80} \right) \right].$$

1. Frequencies at or above 100 Hz were set to zero.
2. The inverse transform was computed, and every downsampling-factor-th sample was retained.
3. The across-channel mean was subtracted at every target-rate sample, implementing an average reference across the selected EEG channels.

This is a frequency-domain antialiasing and decimation procedure applied to the selected 60-second interval. The source contains no separate line-noise filter, bad-channel rejection, or channel interpolation step.

At 250 Hz, each analyzed condition contained 15,000 target-rate samples.

Windowing and artifact screening

Entropy was calculated on nonoverlapping windows. The primary window length was four seconds, producing 15 nominal windows per 60-second acquisition; the sensitivity analysis used two-second windows, producing 30 nominal windows.

Artifact screening was performed separately for each participant, condition, and window length:

1. Peak-to-peak amplitude was calculated within each window and channel.
2. The median peak-to-peak value across channels defined a global value for that window.
3. The median and median absolute deviation (MAD) of these window-level global values were calculated.
4. A window was retained only when its global value was no greater than both 500 μV and the condition-specific median plus six MADs.

The implementation adds 10^{-9} to the MAD before applying the adaptive threshold. It refuses entropy estimation when fewer than $\max(4, \lfloor W/2 \rfloor)$ windows survive, where W is the number of nominal windows.

The participant-level retention and amplitude columns were available to the local run and were checked for finiteness, but those values remain excluded from the manuscript and its figures. Consequently, condition-specific retention distributions and associations with entropy differences cannot be reported here.

Permutation entropy

For each retained window, channel, and scale, order-3 normalized permutation entropy was calculated as

$$H_{\text{PE}} = -\frac{1}{\ln(3!)} \sum_{\pi} p(\pi) \ln p(\pi),$$

where the sum is over observed ordinal patterns and the conventional contribution of an unobserved pattern is $0 \ln 0 = 0$ [3].

All overlapping sequences of three consecutive coarse-grained samples were used, corresponding to embedding order 3 and delay one in coarse-grained samples. Patterns were generated using stable ascending sorting. When samples were tied, the earlier sample index preceded the later index, providing a deterministic

index-based tie rule. The natural logarithm was used in both numerator and normalization.

Coarse-graining

The scales were

$$s \in \{1, 2, 4, 8\}.$$

For each retained window and scale s , the source retained the longest prefix whose length was divisible by s , divided it into nonoverlapping blocks of s samples, and replaced each block with its arithmetic mean. Permutation entropy was then calculated separately for every channel.

Let H_{icwsj} denote entropy for participant i , condition c , retained window w , scale s , and channel j . Window aggregation was

$$h_{icsj} = \text{median}_w H_{icwsj}.$$

Thus, ordinal patterns were not pooled across windows or channels.

Integrated primary endpoint

The per-channel integrated score was the normalized trapezoidal area over base-2 log scale:

$$A_{icj} = \frac{\text{trapz}(h_{icsj}, \log_2 s)}{\log_2 8 - \log_2 1}.$$

Because $\log_2 s = (0, 1, 2, 3)$, this is equivalently

$$A_{icj} = \frac{h_{ic,1,j} + 2h_{ic,2,j} + 2h_{ic,4,j} + h_{ic,8,j}}{6}.$$

The participant-condition endpoint was

$$B_{ic} = \text{median}_j A_{icj}.$$

This formula explains why the endpoint remains on approximately the same zero-to-one scale as normalized permutation entropy. It also clarifies its construct: B_{ic} is a weighted average of entropy level across the selected log-spaced scales, followed by a channel median. It is not an isolated statistic of scale-profile slope, curvature, persistence, or breadth.

The stored participant-level scale columns were computed as $\text{median}_j h_{icsj}$. Applying the quadrature to these median scale columns is not algebraically identical to taking the median of per-channel areas, because median aggregation and integration need not commute.

Regional, window-length, and exploratory endpoints

For the posterior sensitivity analysis, channel names beginning with **O**, **PO**, **P**, or **CP** were classified as posterior. For the frontal analysis, names beginning with **FP**, **AF**, or **F** were classified as frontal. All other channels were excluded from those regional summaries. Regional endpoints were the median per-channel integrated area within the relevant set.

The two-second sensitivity analysis repeated artifact screening, coarse-graining, entropy estimation, window medians, log-scale quadrature, and channel aggregation using two-second rather than four-second windows.

The single-scale endpoint was the median across channels of the four-second-window entropy summary at scale 1.

Spectral outcomes

Spectral metrics were calculated from the complete 60-second target-rate array, not from the retained entropy windows. After per-channel temporal demeaning, the source applied a Hanning taper over the entire interval and calculated squared Fourier magnitudes. Power was median-aggregated across channels at every frequency.

The spectral exponent was the negative slope from an ordinary linear fit of

$$\log_{10} P(f)$$

against $\log_{10} f$ over 2–40 Hz, excluding 8–13 Hz. A constant 10^{-20} was added to power before logarithmic transformation.

Relative alpha power was

$$\frac{\int_8^{13} P(f) df}{\int_2^{40} P(f) df},$$

with both integrals calculated by trapezoidal quadrature on the median spectrum. These definitions make the stored spectral outcomes reproducible, but they do not make them spatially resolved or independent mechanistic controls.

Statistical analysis

Participant-condition differences were defined as sedated minus awake. Subject-level metrics, rather than windows or channels, were the unit of inference.

For every reported metric, the source calculated:

- awake and sedated arithmetic means;
- the mean and median paired difference;
- paired

$$d_z = \frac{\bar{d}}{s_d},$$

where s_d is the sample standard deviation with denominator $n - 1$;

- the number of participants whose difference had the same sign as $\bar{d}$;
- a leave-one-participant-out indicator requiring every deletion estimate to preserve the full-sample mean's sign;
- an exhaustive two-sided sign-flip value; and
- a participant-paired percentile bootstrap interval.

For $n = 20$, the sign-flip calculation enumerated all 2^{20} assignments:

$$p_{\text{SF}} = 2^{-20} \sum_{\epsilon \in \{-1,1\}^{20}} \mathbf{1} \left[\left| \frac{1}{20} \sum_{i=1}^{20} \epsilon_i d_i \right| \geq |\bar{d}| \right].$$

For this paired-mean statistic, sign reversal is mathematically equivalent to swapping condition labels within a pair. Enumeration is computationally exact. An exact randomization interpretation would require randomized exchangeable labels under a sharp null; a distributional interpretation requires joint invariance of the differences under independent sign changes, commonly motivated by symmetry. Neither design-based exchangeability nor distributional symmetry was established here.

For each metric, the bootstrap sampled 20 participant differences with replacement, calculated their mean, and repeated this process 10,000 times. The reported interval comprises the empirical 2.5th and 97.5th percentiles. Pairing was preserved because resampling operated on paired differences. The configured seed was 490213; the full-analysis source initialized its generator with `seed + 1`, or 490214. Metrics shared this generator sequentially, with the primary endpoint processed first.

No multiplicity adjustment, smallest relevant effect, or binary hypothesis-decision threshold was specified.

Results

Completion and post-run consistency verification

The aggregate result recorded 20 completed participant pairs and no failures. A separately written verifier read the frozen participant table and result files and reported that:

- all configured participants completed;
- identifiers were unique;
- all stored numerical values were finite;
- analyzed duration was 60 seconds per acquisition;
- each analyzed recording contained 62 EEG channels;
- physical scaling was 0.1 μV ;
- the primary paired mean difference matched the aggregate result;
- the exhaustive sign-flip value matched;
- the stored endpoint was consistent with the scale columns under the verifier's predefined tolerance;
- the synthetic white-noise entropy exceeded periodic-sequence entropy; and
- the analysis-source and configuration hashes matched the run manifest.

The verifier independently implemented the primary difference, sign enumeration, scale-column calculation, and synthetic entropy check. It did not collect new data, use an independent cohort, rerun acquisition, or provide independent scientific replication. Its role was post-run consistency verification of the same frozen analysis artifacts.

The scale-column check integrated each participant's median-across-channel scale values and required the result to differ from the stored median-of-per-channel-area endpoint by less than 0.02. This is a tolerance-based consistency check, not an assertion of exact equality between two noncommuting aggregation orders.

Protocol-designated primary endpoint

Figure 1 shows the paired data rather than only its aggregate table. Fourteen participants had lower integrated entropy in the selected sedated acquisition and six had higher values. The negative mean is therefore a population summary over heterogeneous paired responses, not a universal participant-level decrease.

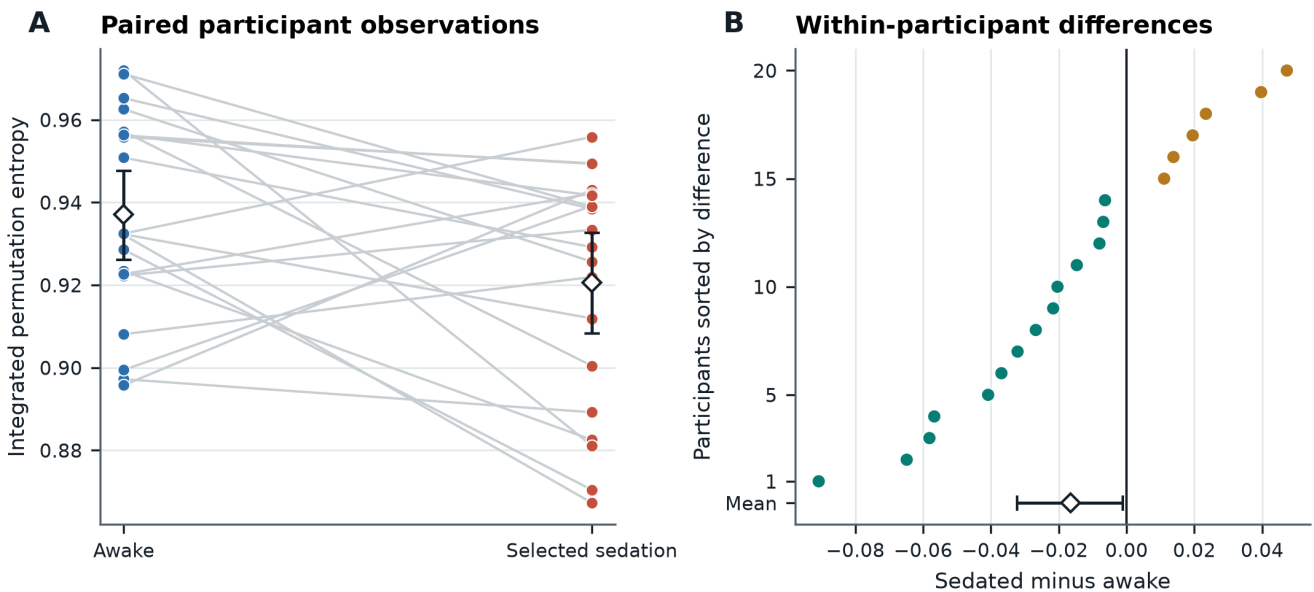


Figure 1. Paired primary-endpoint observations for 20 participants. Panel A connects each participant's awake and selected-sedation integrated permutation-entropy scores; open diamonds and error bars show condition means with participant-bootstrap 95% intervals. Panel B sorts the paired sedated-minus-awake differences without displaying identifiers; teal points are decreases and gold points increases. The open diamond at Mean and its horizontal interval show the mean difference of -0.0165 and the 10,000-resample paired percentile-bootstrap 95% interval of -0.0322 to -0.0011 generated for this figure. Six participants increased, illustrating heterogeneity despite the negative sample mean.

The registered aggregate summary was:

Quantity	Aggregate value
Paired participants	20
Awake mean	0.937074
Sedated mean	0.920556
Mean difference, sedated minus awake	-0.016518
Median difference	-0.017526
Paired d_z	-0.458
Negative differences	14 of 20
Exhaustive two-sided sign-flip p	0.054569
Paired percentile-bootstrap 95% interval	[-0.032088, -0.001146]
Mean sign retained after every leave-one-out omission	Yes

The integrated score was lower on average in the selected sedated acquisition. The mean, median, sign count, bootstrap interval, and leave-one-out indicator summarize the same 20 differences and should not be treated as independent lines of evidence.

The sign-flip and bootstrap summaries need not agree. The sign-flip procedure evaluates the observed mean against sign-reassigned differences under a sign-exchangeability premise. The percentile bootstrap estimates the sampling distribution of the mean by resampling observed participant differences under an independent-participant sampling premise. No rule gave either procedure confirmatory priority.

The leave-one-out result establishes only that deleting any single member of this sample did not reverse the estimated mean’s sign. It does not establish robustness to participant sampling, preprocessing, acquisition choice, artifact thresholds, scale weighting, or estimator parameters.

The local participant table permits the descriptive scale-profile visualization in Figure 2A, but no registered condition-by-scale interaction or curve-shape model was supplied. The plot can reveal the observed profile geometry without converting it into a confirmatory breadth test.

Window-length and regional sensitivity analyses

Figure 2A exposes the scale profiles underlying the scalar endpoint. Both condition means rise with coarse-graining scale, and the largest visible separation occurs at scale 1; the curves do not by themselves identify a change in profile breadth. Figure 2B places the primary, window-length, regional, and single-scale estimates on the same raw entropy-difference axis. The posterior estimate is small and uncertain, whereas the frontal and scale-1 estimates are more negative.

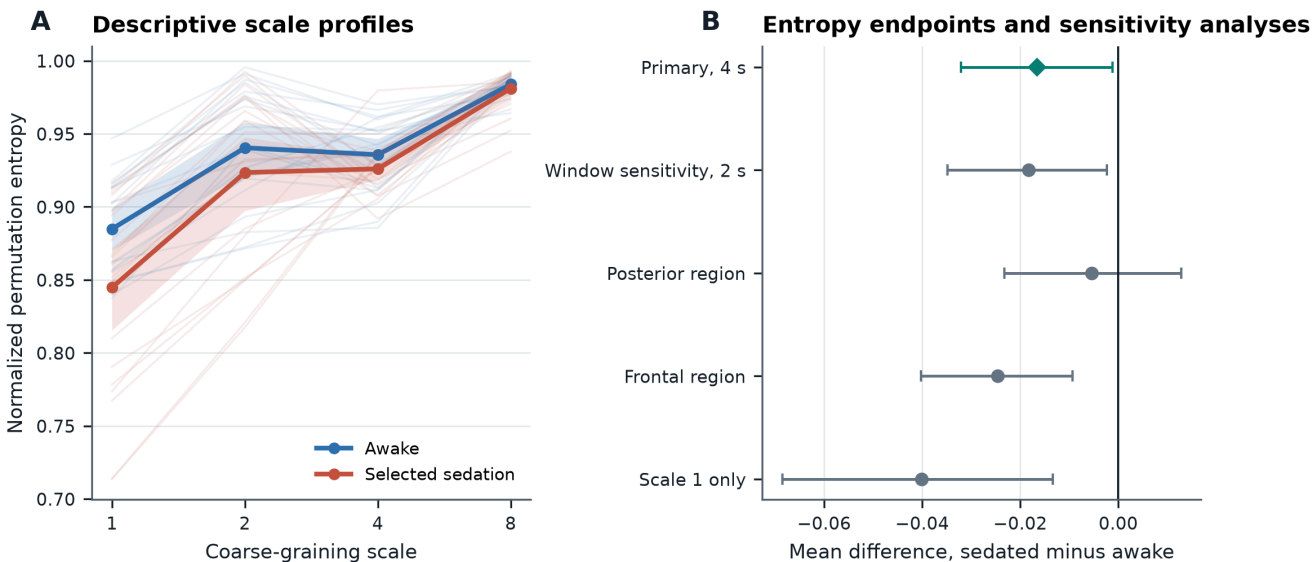


Figure 2. Scale profiles and sensitivity analyses. Panel A shows participant-level scale profiles as faint lines and condition means with pointwise participant-bootstrap 95% intervals as thick lines. These descriptive curves do not constitute a registered condition-by-scale interaction test. Panel B shows stored mean sedated-minus-awake differences and paired percentile-bootstrap 95% intervals for the primary endpoint and four related entropy summaries. The posterior interval crosses zero; the analyses are correlated and no multiplicity correction or joint decision rule was registered.

The exact sensitivity summaries were:

Outcome	Awake mean	Sedated mean	Mean difference	Median difference	d_z	Sign-flip p	Bootstrap 95% interval	Same-sign participants
Two-second integrated entropy	0.933434	0.915140	-0.018293	-0.018325	-0.475	0.047562	[-0.034804, -0.002279]	14

Outcome	Awake mean	Sedated mean	Mean difference	Median difference	d_z	Sign-flip p	Bootstrap 95% interval	Same-sign participants
Posterior integrated entropy	0.926376	0.920975	-0.005401	-0.007455	-0.127	0.574911	[-0.023226, 0.012787]	13
Frontal integrated entropy	0.939393	0.914773	-0.024620	-0.025281	-0.677	0.006826	[-0.040263, -0.009342]	15

All three leave-one-participant-out indicators preserved their respective full-sample mean signs.

The two-second estimate was similar to the four-second estimate. The frontal estimate was more negative, whereas the posterior estimate was small and its interval included both decreases and increases.

The literal protocol criterion that the direction survive window-length and regional sensitivities was met at the level of point-estimate signs. This is a weak descriptive criterion: an arbitrarily small negative posterior estimate qualifies even when its uncertainty includes effects in both directions. The analyses therefore do not establish spatial uniformity or meaningful regional robustness.

Pilot and full-analysis structure

The configuration designated four participants for `--pilot` pipeline execution. The normal full-analysis command used all 20 configured participants, including those four. The aggregate package did not contain a pilot-only estimate or a nonpilot-only estimate.

The full result is therefore a pooled analysis, not an independent confirmation in 16 held-out participants. The artifacts also do not establish whether any settings changed after pilot inspection.

Robustness and Negative Controls

Single-scale entropy

Quantity	Aggregate value
Awake mean	0.884848
Sedated mean	0.844783
Mean difference	-0.040065
Median difference	-0.011156
Paired d_z	-0.623
Sign-flip p	0.010935
Bootstrap 95% interval	[-0.068515, -0.013387]
Same-sign participants	13 of 20
Leave-one-out sign retained	Yes

The scale-1 result supports a lower fine-scale entropy level in the selected sedated acquisition. It also makes a general entropy-level shift a plausible explanation for the integrated difference. Without aggregate contrasts for every scale or a condition-by-scale model, it cannot establish altered scale dependence.

Contextual spectral outcomes

Outcome	Awake mean	Sedated mean	Mean difference	Median difference	d_z	Sign-flip p	Bootstrap 95% interval	Same-sign participants
Spectral exponent	1.573989	2.057534	0.483544	0.559043	0.921	0.001040	[0.243631, 0.698746]	17
Relative alpha power	0.445180	0.344708	-0.100472	-0.124145	-0.621	0.013029	[-0.169801, -0.031752]	14

Both leave-one-participant-out indicators preserved their mean directions.

These results document accompanying differences in the defined whole-scalp spectral summaries. Relative alpha here specifically means 8–13-Hz power divided by 2–40-Hz power in the median channel spectrum; it is not a topographically resolved alpha measure.

Permutation entropy can change as autocorrelation and spectral composition change. No phase-randomized, spectrum-preserving, spectrum-matched, or amplitude-distribution-preserving surrogate analysis was implemented. The available results therefore do not show that the entropy difference contains information beyond conventional spectral or autocorrelation differences.

Synthetic estimator check

The post-run verifier generated:

- 20,000 standard-normal samples using `numpy.random.default_rng(17)`; and
- a 20,000-sample periodic sequence formed by repeating `[0, 1, 0, -1]` 5,000 times.

It independently calculated order-3, unit-delay normalized permutation entropy using stable sorting. Entropy was 0.999869 for white noise and 0.773706 for the periodic sequence. The expected ordering was therefore reproduced.

This is a useful estimator sanity check, but it is not a negative control for the biological comparison, a test of spectral specificity, or validation of the endpoint as a consciousness measure.

Artifact and effective-sample limitations

The artifact algorithm and minimum-retention rule are fully specified in the supplied source. However, participant-by-condition retention counts and median peak-to-peak values were intentionally withheld from the aggregate manuscript package. The following analyses were not reported:

- matching the number of retained windows across conditions;
- associations between entropy differences and retention differences;
- associations between entropy and peak-to-peak amplitude;
- alternative absolute or MAD-based artifact thresholds; and

- amplitude- or artifact-matched comparisons.

Artifact selection and effective-sample differences therefore remain possible contributors.

Controls not performed

The supplied source contained no:

- randomized acquisition-order control;
- sham-sedation condition;
- alternate awake- or sedated-segment analysis;
- matched equal-window analysis;
- spectrum-preserving or IAAFT surrogate analysis;
- amplitude-matched analysis;
- condition-by-scale interaction or level-shift comparison;
- alternative permutation order or embedding delay;
- alternative coarse-graining rule or integration weighting;
- nonpilot-only confirmatory analysis; or
- independent replication dataset.

The exhaustive sign-flip computation is already equivalent to within-pair label swapping for the mean-difference statistic. Its limitation is the absence of a design or substantiated distributional assumption that makes those swaps exchangeable.

Interpretation

The selected sedated acquisition had lower log-scale-integrated permutation entropy than the selected eyes-closed awake acquisition. The four-second mean difference was -0.016518 , and the two-second analysis yielded a similar estimate of -0.018293 .

The methods underlying that scalar are now fully specified by the supplied source. It is a normalized trapezoidal average over $\log_2$ scale after median aggregation across retained windows within channels, followed by a channel median. Accordingly, the finding is not merely an uninterpretable stored number.

Its construct interpretation nevertheless remains limited. The scalar quantifies integrated entropy level over four chosen scales. It does not isolate breadth, persistence, slope, curvature, or a condition-by-scale interaction. The comparatively large single-scale difference is compatible with a general entropy-level shift. Figure 2A makes a general level shift visually plausible, but without a registered paired condition-by-scale model that possibility cannot be separated inferentially from a change in scale-profile shape.

The statistical evidence is mixed across procedures. The paired percentile-bootstrap interval excluded zero, whereas the sign-flip value was 0.054569 . The sign-flip enumeration was computationally exact but depends inferentially on sign exchangeability. The bootstrap was fully specified and preserved pairing, but it rests on participant-sampling assumptions and was not paired with a prespecified decision threshold. Neither result warrants a confirmatory declaration.

The protocol's directional sensitivity statement was met literally: the two-second, posterior, and frontal point estimates were negative. The posterior evidence was weak, with a small standardized estimate and an interval spanning both directions. Directional agreement among correlated summaries does not constitute independent convergence.

The comparison also does not identify a propofol-specific causal effect. The conditions were not shown to be randomized or exchangeable, and they may differ in order, recording context, vigilance, fatigue, preceding events, and pharmacological state. The evidence supports an association between the selected acquisition and the integrated entropy score.

Finally, the accompanying spectral differences and absence of spectrum-preserving surrogates prevent interpretation of the entropy contrast as a spectral-independent nonlinear dynamical effect.

Relevance to Consciousness Research

Neural complexity provides the motivation for this analysis, but consciousness was not its measured outcome. Perturbational complexity was designed to characterize cortical responses to controlled stimulation and is methodologically distinct from passive permutation entropy ^[6]. The present statistic was not validated against reports, responsiveness, or another measure of conscious state.

The source study concerned repeated awakenings and dreaming during propofol sedation ^[1]. That context does not convert an acquisition label into an experience label. The supplied protocol and aggregate record state that the analyzed event tables exposed no dream-report labels, and no such labels were reconstructed ^[7].

The analysis therefore cannot determine whether a participant was conscious, unconscious, dreaming, or without experience during the selected segment. Sedation and phenomenal unconsciousness are not equivalent constructs. The result cannot establish a neural correlate of consciousness, discriminate among theories of consciousness, validate a clinical consciousness marker, or classify individual awareness.

Prior reports of reduced spontaneous-EEG complexity during propofol general anesthesia provide contextual consistency only ^[5]. They do not show that the present integrated permutation-entropy score measures the same construct or that either measure directly indexes phenomenal experience.

Limitations

First, temporal preregistration was not independently established. The supplied README calls the hypothesis and full analysis registered, and the configuration and source provide a complete computational specification, but no independently time-stamped protocol or version history was supplied.

Second, the endpoint is a weighted average of entropy levels, not a unique measure of multiscale breadth. This revision adds descriptive scale curves, but registered interaction models and shape comparisons against a common level shift remain unavailable.

Third, the four pilot-designated participants were included in the full $n = 20$ analysis. No pilot-only, nonpilot-only, or setting-selection audit was supplied, so held-out confirmation cannot be claimed.

Fourth, no smallest relevant effect, alpha threshold, multiplicity procedure, regional uncertainty requirement, or joint success rule accompanied H2. Its directional sensitivity criterion is satisfied by any negative point estimate, regardless of magnitude or precision.

Fifth, sign-flip inference requires randomized label exchangeability or an appropriate sign-symmetry assumption. Neither was established. The paired percentile bootstrap is reproducible but relies on the observed participant sample as an empirical approximation to a broader sampling distribution.

Sixth, participant identifiers and participant-level rows remain withheld from the public manuscript package. Figure 1 now exposes the anonymized difference distribution and its asymmetry, but readers still cannot map points to acquisition metadata, recompute alternative intervals from published rows, or audit influential observations directly.

Seventh, the nonrandomized acquisition contrast is causally ambiguous. Dose, responsiveness, physiological covariates, and quantitative sedation depth were unavailable. Pairing does not remove order, drift, vigilance, fatigue, preceding-event, or context effects.

Eighth, the code analyzes the first 60 seconds of each selected acquisition. This rule is explicit, but alternate segments were not examined, and independently time-stamped evidence that the rule preceded outcome inspection was not supplied.

Ninth, artifact screening can retain different windows across conditions. Participant-level retention and amplitude summaries were not included in the aggregate package, and no equal-window, amplitude-matched, threshold-sensitivity, or retention-association analysis was reported.

Tenth, no separate bad-channel removal, interpolation, or line-noise procedure was implemented. The frequency-domain filter was applied only to the selected 60-second interval, so its behavior also depends on the finite-interval Fourier representation.

Eleventh, no spectrum-preserving or spectrum-matched surrogate analysis was performed. The entropy difference cannot be separated from spectral, autocorrelation, or noise-mixture changes.

Twelfth, regional evidence was heterogeneous. The frontal estimate was appreciably negative, while the posterior estimate was small and uncertain. The prefix-based channel definitions are reproducible but coarse and were not accompanied by topographic modeling.

Thirteenth, the spectral outcomes were whole-scalp summaries. They describe the median channel spectrum but cannot establish spatially specific physiological effects.

Fourteenth, the synthetic white-noise/periodic-sequence comparison verifies an expected estimator ordering only. It supplies neither biological specificity nor a null distribution for the acquisition contrast.

Fifteenth, novelty relative to the companion publication was not established by a method-by-method comparison. The supplied evidence does not determine whether that publication analyzed overlapping segments, permutation entropy, or a related multiscale endpoint. No novelty claim is made here.

Sixteenth, the local result interprets the dataset README as CC BY 4.0 while noting that derivative metadata report CCo. The present package does not resolve which notice governs each artifact.

Finally, no report of experience, dreaming, awareness, or absence of experience entered the analysis. No conclusion about phenomenal consciousness follows from the observed EEG association.

Reproducibility

The public source dataset is OpenNeuro ds005620 version 1.0.0, DOI 10.18112/openneuro.ds005620.v1.0.0 [2]. The supplied local artifact collection includes the protocol README, sanitized configuration, aggregate result JSON, complete analysis source, verifier source, and verifier output [7].

The protocol specifies the following commands:

```
.venv/bin/python experiments/sabine-laghari/propofol-multiscale-dynamics/analyze.py --pilot
.venv/bin/python experiments/sabine-laghari/propofol-multiscale-dynamics/analyze.py
```

The first runs the four-participant pipeline pilot; the second runs the full configured sample.

Parameter	Supplied value
Dataset	OpenNeuro ds005620 v1.0.0
Conditions	task-awake_acq-EC ; task-sed2_acq-rest_run-1
Paired participants completed	20
Configured excluded participants	1
Pipeline-pilot participants	4
Analyzed duration	First 60 seconds per acquisition
Verified EEG channels	62
Target sampling rate	250 Hz
Target samples per condition	15,000
Verified physical scale	0.1 µV
Antialias pass/stop frequencies	80/100 Hz
Primary window length	4 seconds
Window sensitivity	2 seconds
Permutation order and delay	3; one coarse-grained sample
Tie rule	Stable sort; earlier index precedes later index
Coarse-graining	Nonoverlapping block means
Scales	1, 2, 4, 8
Integration axis	log ₂ scale
Channel aggregation	Median of per-channel areas
Absolute artifact threshold	500 µV
Adaptive artifact threshold	Median + 6 MAD

Parameter	Supplied value
Bootstrap resamples	10,000 participant-paired resamples
Bootstrap interval	Percentile, 2.5th–97.5th percentiles
Configured seed	490213
Full-analysis generator seed	490214
Primary endpoint	<code>multiscale_entropy_auc</code>

The analysis source writes a manifest containing the command, Python and NumPy versions, platform, seed, source and configuration SHA-256 hashes, raw-file paths, sizes and hashes, and output-file sizes and hashes. The post-run verifier confirmed that the supplied analysis source and configuration matched their manifest hashes, although the digest strings and participant-level manifest rows are not reproduced here.

From the frozen participant table, the verifier independently recomputed the exact primary mean difference,

$$-0.016518164674441033,$$

and exhaustive two-sided sign-flip value,

$$0.054569244384765625.$$

It also checked all configured completions, identifier uniqueness, finiteness, duration, channel count, physical scaling, and consistency between stored scale columns and the endpoint. These checks support the computational consistency of the frozen run. They are not independent scientific replication because they use the same participants, acquisitions, and stored outputs.

Exact external reconstruction of the reported cohort from this manuscript alone is limited because participant identifiers and participant-level rows were intentionally withheld. Reanalysis also requires access to the unsanitized subject roster and its exclusion rationale.

The aggregate result describes the dataset README license as CC BY 4.0 and notes CCo in derivative metadata. Reusers should verify the governing notice for each file in the versioned OpenNeuro repository.

Conclusion

Among 20 paired participants, log-scale-integrated permutation entropy was lower on average in the selected propofol-sedation acquisition than in the selected eyes-closed awake acquisition. The paired mean difference was -0.016518 . Two-second, posterior, and frontal point estimates were also negative, satisfying the protocol’s literal directional sensitivity criterion, although the posterior estimate was small and uncertain.

The supplied analysis and verifier resolve the previous uncertainty about coarse-graining, embedding delay, tie handling, aggregation order, AUC weighting, filtering, regional definitions, bootstrap construction, synthetic inputs, and post-run checks. The endpoint is reproducibly defined, and its central arithmetic was independently reimplemented and verified against the frozen artifacts.

Those improvements do not remove the study's inferential and construct limitations. The scalar measures a weighted average of entropy across selected log-spaced scales, not scale-profile breadth or shape. The acquisition labels were not shown to be randomized or sign-exchangeable, temporal preregistration was not independently established, pilot participants were included in the pooled analysis, and no confirmatory decision rule was specified.

The result also does not identify a propofol-specific causal effect or a spectral-independent mechanism. It describes an association between two selected acquisitions that may include pharmacological state, order, context, vigilance, artifact selection, and spectral differences.

Finally, this was not a dream-report or awareness analysis. It provides no evidence that any participant was phenomenally conscious, unconscious, dreaming, or without experience.

References

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